SeedingMinds

Assessment Name

Java_S3_271225

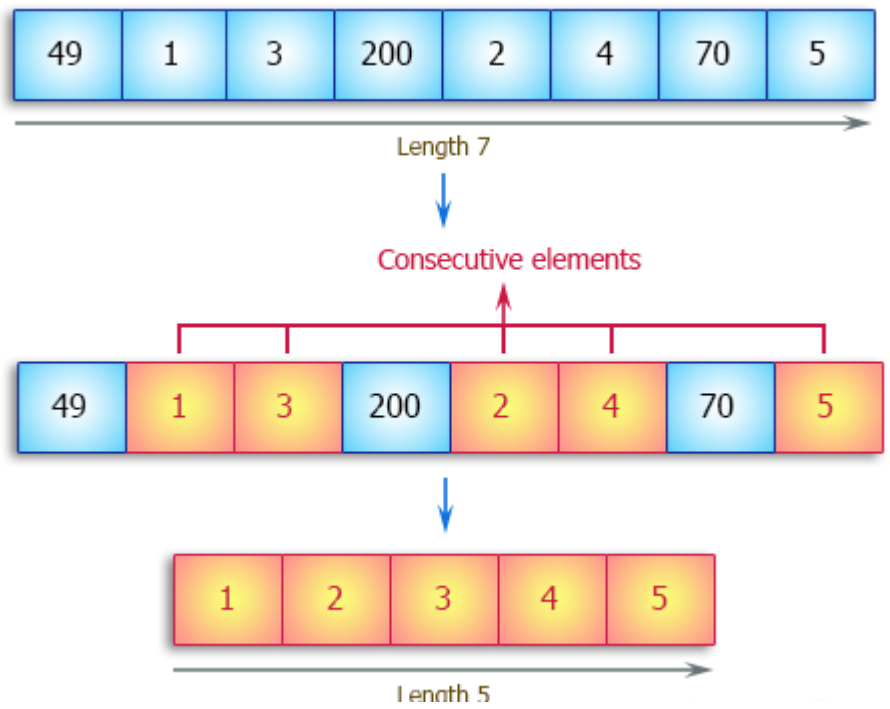
Module: Coding

Question 1:

Marks: +10 -0

length of the longest consecutive elements sequence from a given unsorted array of integers

Write a Java program to find the length of the longest consecutive elements sequence from a given unsorted array of integers.
Sample array: [49, 1, 3, 200, 2, 4, 70, 5]
The longest consecutive elements sequence is [1, 2, 3, 4, 5], therefore the program will return its length 5.



Constraints:

N/A

Example:

Input:

49 1 3 200 2 4 70 5

Output:

5

Explanation:

-

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	49 1 3 200 2 4 70 5	5

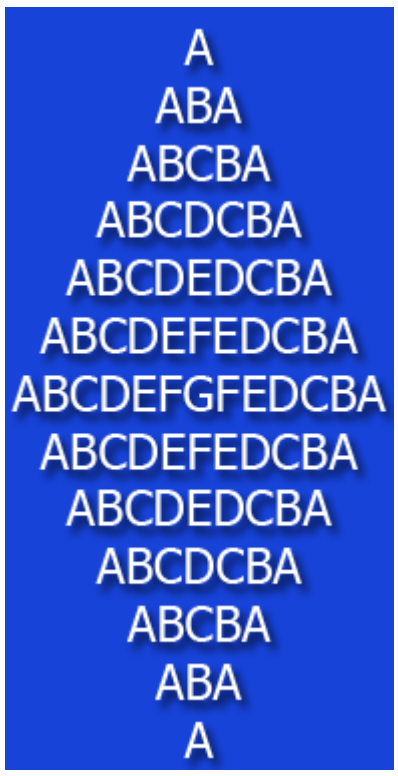
Question 2:

Marks: +10 -0

display the following character rhombus structure.

Write a Java program to display the following character rhombus structure.

Pictorial Presentation:



Constraints:

N/A

Example:

Input:

7

Output:

```
      A
     ABA
    ABCBA
   ABCDCBA
  ABCDEDCBA
 ABCDEFEDCBA
ABCDEFGFEDCBA
 ABCDEFEDCBA
  ABCDEDCBA
   ABCDCBA
    ABCBA
     ABA
      A
```

Explanation:

-

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	7	A ABA ABCBA ABCDCBA ABCDEDCBA ABCDEFEDCBA ABCDEFGFEDCBA ABCDEFEDCBA ABCDEDCBA ABCDCBA ABCBA ABA A

Question 3:

Marks: +10 -0

Electricity Bill Calculator - 2

Write a program to calculate and print the Electricity bill of a given customer. The unit consumed by the user should be taken from the keyboard and display the total amount to pay to the customer.

The charge are as follow -

Unit	Charge/unit
upto 199	@1.20
200 and above but less than 400	@1.50
400 and above but less than 600	@1.80
600 and above	@2.00

Conditions:

a. If bill exceeds Rs.300 then a surcharge of 15%(On the bill amount) will be added on the total bill to pay

b. If the bill amount is less than Rs.100, then the minimumto be paid will be of Rs.100/-

Note: Round off the result to 2 decimal points.

Constraints:

Input Format:

Line 1: Integer Number (Number of units used/consumed)

Output Format:

Line 1: Floating-point number (Bill amount)

Line 2 : Floating-point number (Surcharge amount if any otherwise print as 0.00)

Line 3: Floating-point number (Net amount Payable - Bill amount + Surcharge amount)

Example:

Input:

320

Output:

480.00

72.00

552.00

Explanation:

Explanation:

Total units consumed is 320

For that Bill amount to pay will come as Rs.480 which is greater than Rs.300

Hence Surcharge of 15% will be collected on the Bill amount, which = Rs.72

Net Payable amount = Rs.552 (480+72)

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	320	480.00 72.00 552.00

#	INPUT	EXPECTED OUTPUT
2	900	1800.00 270.00 2070.00

Question 4:

Marks: +10 -0

Text Mirror Comparator

A social media moderation tool allows users to post mirrored text for artistic or creative effects.
Before publishing, the system must **check if a user's text is identical to its reversed version** (i.e., a palindrome-like validation used for symmetric text checking).

You are required to:

1. Take a string as input.
2. Reverse the string manually (without using string functions).
3. Display both the **original** and **reversed** strings.
4. If both are identical (case-insensitive), print `"Mirror Text"` .
   Otherwise, print `"Not Mirror Text"` .

This concept is used in platforms that validate or generate symmetric hashtags, mirror art names, or reflective usernames.

Constraints:

- Input string length ≤ 100.
- Use only arrays, control, and conditional statements.
- No pointers, no built-in string functions.
- Ignore case during comparison (i.e., "Level" = "level").

Example:

Sample Input:

Level

Sample Output:

Original: Level
Reversed: leveL
Mirror Text

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	radar	Original: radar Reversed: radar Mirror Text